

COURSE CODE/TITLE: CPE 212 Automating Server Management	SECTION: CpE31S2
DATE PERFORMED: 07/09/2026	DATE SUBMITTED: 07/09/2026
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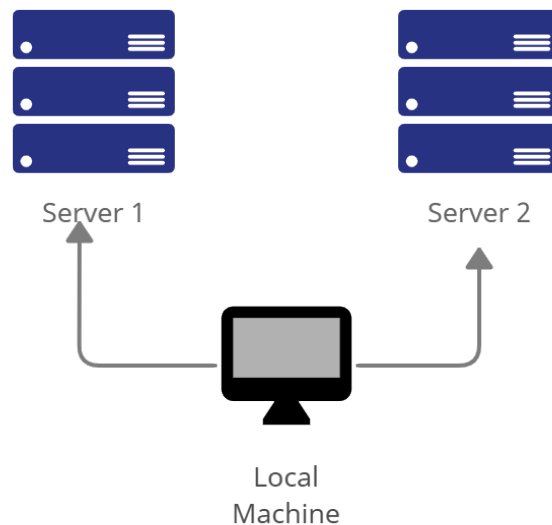
ACTIVITY NO. ____ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

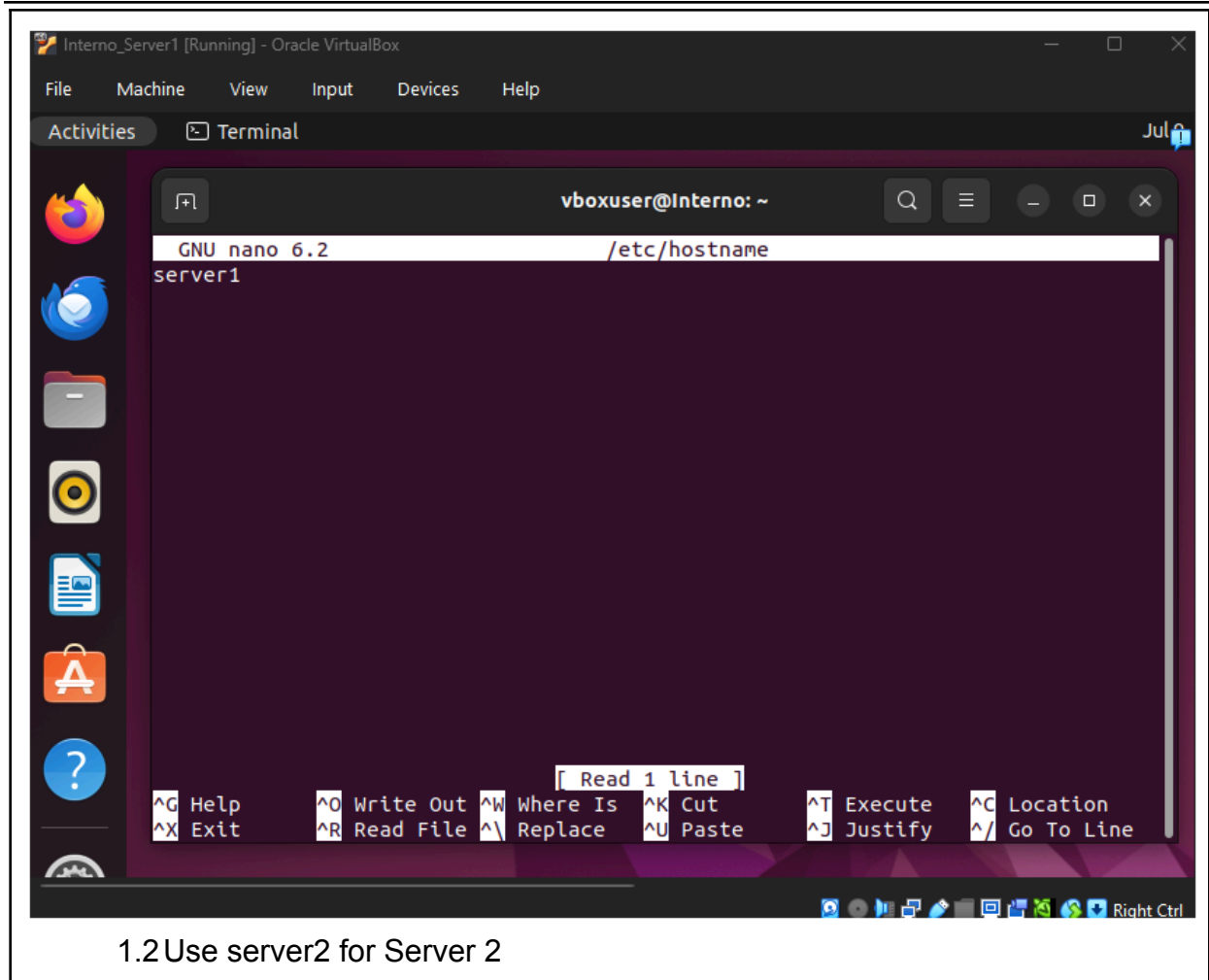
Network Topology:

Assume that you have created the following network topology in Virtual Machines, *provide screenshots for each task*. (Note: *it is assumed that you have prior knowledge of cloning and creating snapshots in a virtual machine*).

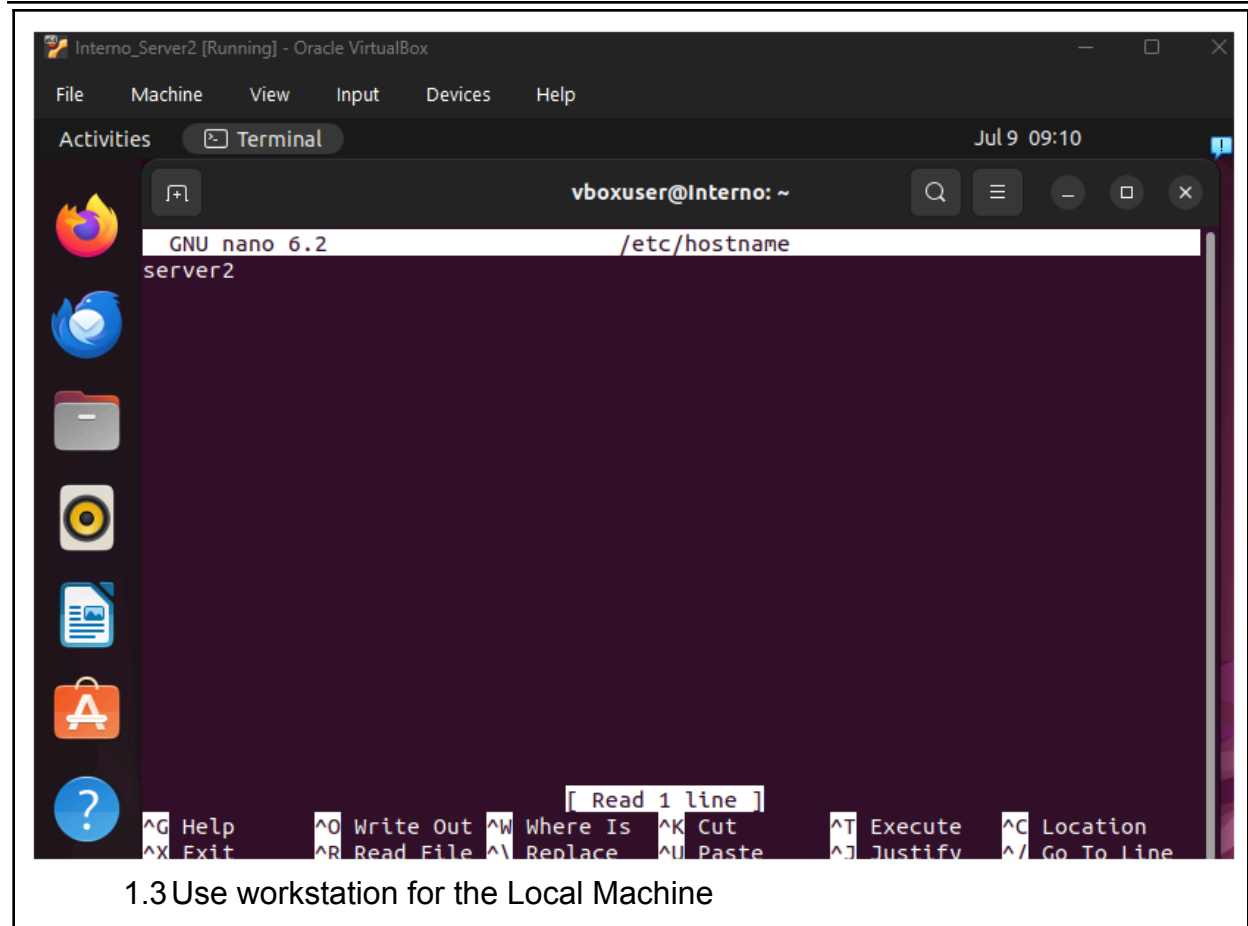


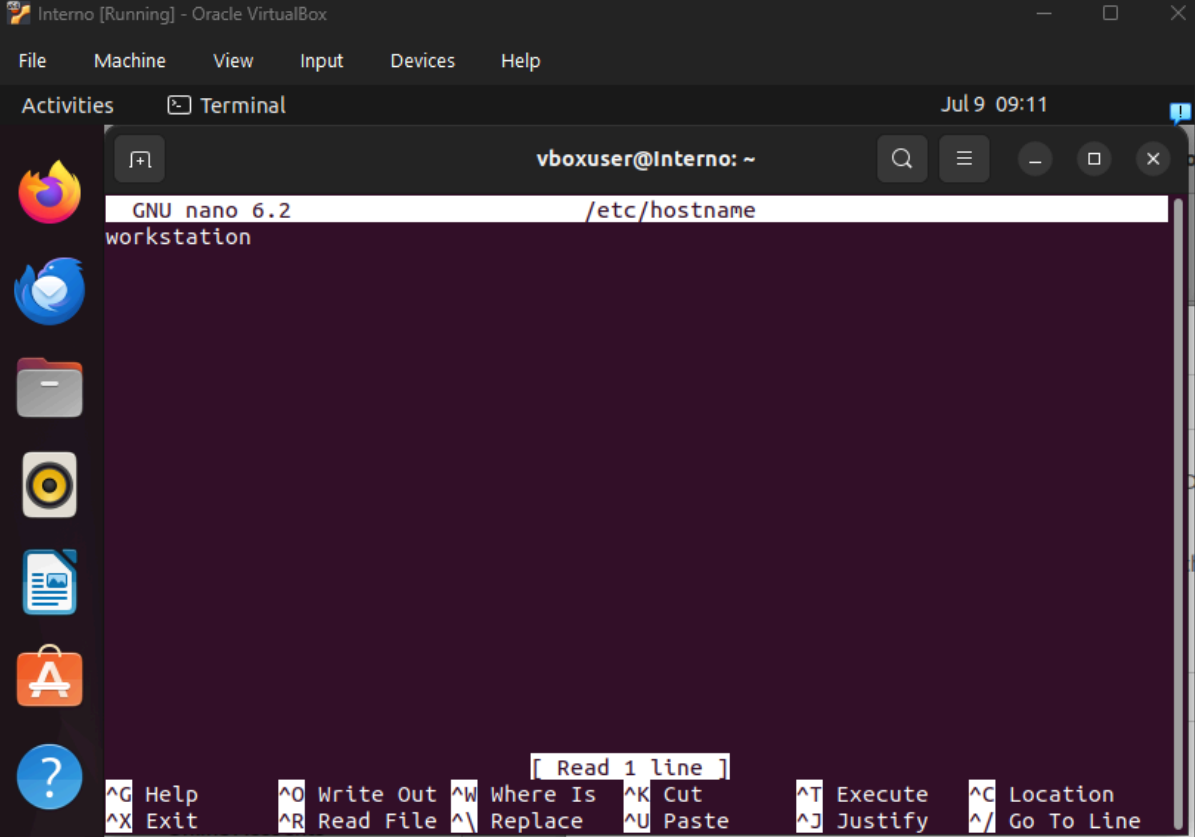
Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command *sudo nano /etc/hostname*
 - 1.1 Use server1 for Server 1



1.2 Use server2 for Server 2





Interno [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Jul 9 09:11

vboxuser@Interno: ~

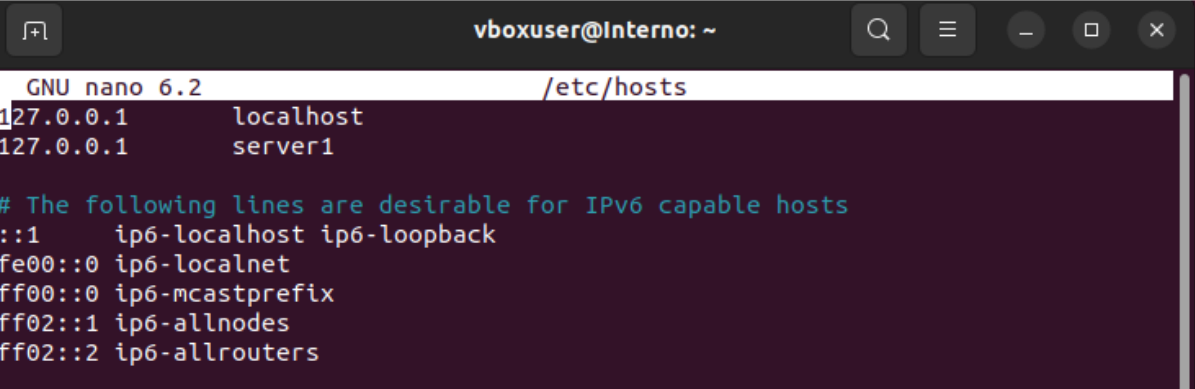
GNU nano 6.2 /etc/hostname

workstation

[Read 1 line]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

2. Edit the hosts using the command *sudo nano /etc/hosts*. Edit the second line.
2.1 Type 127.0.0.1 server 1 for Server 1



vboxuser@Interno: ~

GNU nano 6.2 /etc/hosts

```
127.0.0.1 localhost
127.0.0.1 server1

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

- 2.2 Type 127.0.0.1 server 2 for Server 2

```
vboxuser@Interno: ~
GNU nano 6.2 /etc/hosts
127.0.0.1    localhost
127.0.0.1    server2

# The following lines are desirable for IPv6 capable hosts
::1        ip6-localhost ip6-loopback
fe00::0    ip6-localnet
ff00::0    ip6-mcastprefix
ff02::1    ip6-allnodes
ff02::2    ip6-allrouters
```

2.3 Type 127.0.0.1 workstation for the Local Machine

```
vboxuser@Interno: ~
GNU nano 6.2 /etc/hosts
127.0.0.1    localhost
127.0.0.1    workstation

# The following lines are desirable for IPv6 capable hosts
::1        ip6-localhost ip6-loopback
fe00::0    ip6-localnet
ff00::0    ip6-mcastprefix
ff02::1    ip6-allnodes
ff02::2    ip6-allrouters
```

Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command *sudo apt update* and *sudo apt upgrade* respectively.

server1

```
vboxuser@Interno: ~
vboxuser@Interno:~$ sudo apt update
[sudo] password for vboxuser:
Hit:1 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
595 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

server2

```
vboxuser@Interno:~$ sudo apt update
[sudo] password for vboxuser:
Hit:1 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
595 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

workstation

```
vboxuser@Interno:~$ sudo apt update
[sudo] password for vboxuser:
Hit:1 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
595 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

2. Install the SSH server using the command *sudo apt install openssh-server*.

server1

```
vboxuser@Interno:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 594 not upgraded.
Need to get 1,656 kB of archives.
After this operation, 6,065 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-clien
t amd64 1:8.9p1-3ubuntu0.15 [904 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-
```

server2

```
vboxuser@Interno:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 594 not upgraded.
Need to get 1,656 kB of archives.
After this operation, 6,065 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-clien
t amd64 1:8.9p1-3ubuntu0.15 [904 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-
```

workstation

```
vboxuser@Interno:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 594 not upgraded.
Need to get 1,656 kB of archives.
After this operation, 6,065 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-clien
t amd64 1:8.9p1-3ubuntu0.15 [904 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-
```

3. Verify if the SSH service has started by issuing the following commands:

3.1 *sudo service ssh start*

server1

```
vboxuser@Interno:~$ sudo service ssh start
```

server2

```
vboxuser@Interno:~$ sudo service ssh start
```

workstation

```
vboxuser@Interno:~$ sudo service ssh start
```

3.2 *sudo systemctl status ssh*

server1

```
vboxuser@Interno:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: en>
   Active: active (running) since Thu 2026-07-09 08:57:26 CST; 24min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 6483 (sshd)
     Tasks: 1 (limit: 4609)
    Memory: 1.7M
       CPU: 14ms
    CGroup: /system.slice/ssh.service
           └─6483 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 09 08:57:26 Interno systemd[1]: Starting OpenBSD Secure Shell server...
Jul 09 08:57:26 Interno sshd[6483]: Server listening on 0.0.0.0 port 22.
Jul 09 08:57:26 Interno sshd[6483]: Server listening on :: port 22.
Jul 09 08:57:26 Interno systemd[1]: Started OpenBSD Secure Shell server.
lines 1-16/16 (END)
```

server2

```
vboxuser@Interno:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: en>
   Active: active (running) since Thu 2026-07-09 08:57:26 CST; 24min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 6483 (sshd)
     Tasks: 1 (limit: 4609)
    Memory: 1.7M
       CPU: 14ms
    CGroup: /system.slice/ssh.service
           └─6483 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 09 08:57:26 Interno systemd[1]: Starting OpenBSD Secure Shell server...
Jul 09 08:57:26 Interno sshd[6483]: Server listening on 0.0.0.0 port 22.
Jul 09 08:57:26 Interno sshd[6483]: Server listening on :: port 22.
Jul 09 08:57:26 Interno systemd[1]: Started OpenBSD Secure Shell server.
lines 1-16/16 (END)
```

workstation


```
vboxuser@Interno:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e
   Active: active (running) since Thu 2026-07-09 08:57:26 CST; 24min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 6483 (sshd)
      Tasks: 1 (limit: 4609)
     Memory: 1.7M
        CPU: 14ms
    CGroup: /system.slice/ssh.service
            └─6483 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 09 08:57:26 Interno systemd[1]: Starting OpenBSD Secure Shell server...
Jul 09 08:57:26 Interno sshd[6483]: Server listening on 0.0.0.0 port 22.
Jul 09 08:57:26 Interno sshd[6483]: Server listening on :: port 22.
Jul 09 08:57:26 Interno systemd[1]: Started OpenBSD Secure Shell server.
lines 1-16/16 (END)
```

4. Configure the firewall to all port 22 by issuing the following commands:

4.1 *sudo ufw allow ssh*

server1

```
vboxuser@JeraldServer1:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

server2

```
vboxuser@Jeraldserver2:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

workstation

```
vboxuser@Interno:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

4.2 *sudo ufw enable*

server1

```
vboxuser@JeraldServer1:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

server2

```
vboxuser@Jeraldserver2:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

workstation

```
vboxuser@Interno:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

4.3 *sudo ufw status*

server1

```
vboxuser@JeraldServer1:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
```

server2

```
vboxuser@Jeraldserver2:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
```

workstation

```
vboxuser@Interno:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
1.1 Server 1 IP address: 192.168.56.114

```
vboxuser@JeraldServer1:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.114 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::4601:5937:5077:c2e8 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:5a8c:1eec:23c8:30f0 prefixlen 64 scopeid 0x0<gl
obal>
    inet6 fd17:625c:f037:2:e0f:84a4:6457:52be prefixlen 64 scopeid 0x0<glo
bal>
    ether 08:00:27:56:37:cf txqueuelen 1000 (Ethernet)
    RX packets 61122 bytes 73952760 (73.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 13615 bytes 896145 (896.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 34379 bytes 2745697 (2.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34379 bytes 2745697 (2.7 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

1.2 Server 2 IP address: 192.168.56.115

```
vboxuser@Jeraldserver2:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.115 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fd17:625c:f037:2:aa35:a422:5c4e:a082 prefixlen 64 scopeid 0x0<gl
obal>
    inet6 fe80::8040:5e75:7ba7:16f prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:8787:9ed4:ce36:79f prefixlen 64 scopeid 0x0<glo
bal>
    ether 08:00:27:f3:21:64 txqueuelen 1000 (Ethernet)
    RX packets 3838 bytes 3354341 (3.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2562 bytes 212720 (212.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 2491 bytes 201355 (201.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2491 bytes 201355 (201.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

1.3 Local Machine IP address: 192.168.56.113

```
vboxuser@workstation:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.113 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::6148:689a:ee8d:1883 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:236b:9166:7040:7a36 prefixlen 64 scopeid 0x0<gl
obal>
    inet6 fd17:625c:f037:2:b5a9:3e7e:7f94:48f prefixlen 64 scopeid 0x0<glo
bal>
    ether 08:00:27:8c:a0:9b txqueuelen 1000 (Ethernet)
    RX packets 961 bytes 226555 (226.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 372 bytes 46444 (46.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 285 bytes 24070 (24.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 285 bytes 24070 (24.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: / Successful ☐ Not Successful

```
vboxuser@workstation:~$ ping 192.168.56.114
PING 192.168.56.114 (192.168.56.114) 56(84) bytes of data.
64 bytes from 192.168.56.114: icmp_seq=1 ttl=64 time=1.87 ms
64 bytes from 192.168.56.114: icmp_seq=2 ttl=64 time=0.410 ms
64 bytes from 192.168.56.114: icmp_seq=3 ttl=64 time=0.730 ms
64 bytes from 192.168.56.114: icmp_seq=4 ttl=64 time=0.546 ms
64 bytes from 192.168.56.114: icmp_seq=5 ttl=64 time=0.445 ms
64 bytes from 192.168.56.114: icmp_seq=6 ttl=64 time=0.669 ms
64 bytes from 192.168.56.114: icmp_seq=7 ttl=64 time=0.758 ms
64 bytes from 192.168.56.114: icmp_seq=8 ttl=64 time=0.484 ms
64 bytes from 192.168.56.114: icmp_seq=9 ttl=64 time=1.07 ms
^C
--- 192.168.56.114 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8190ms
rtt min/avg/max/mdev = 0.410/0.775/1.869/0.431 ms
```

2.2 Connectivity test for Local Machine 1 to Server 2: / Successful ☐ Not Successful

```
vboxuser@workstation:~$ ping 192.168.56.115
PING 192.168.56.115 (192.168.56.115) 56(84) bytes of data.
64 bytes from 192.168.56.115: icmp_seq=1 ttl=64 time=0.568 ms
64 bytes from 192.168.56.115: icmp_seq=2 ttl=64 time=0.636 ms
64 bytes from 192.168.56.115: icmp_seq=3 ttl=64 time=0.676 ms
64 bytes from 192.168.56.115: icmp_seq=4 ttl=64 time=0.583 ms
64 bytes from 192.168.56.115: icmp_seq=5 ttl=64 time=0.385 ms
64 bytes from 192.168.56.115: icmp_seq=6 ttl=64 time=0.439 ms
^C
--- 192.168.56.115 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5143ms
rtt min/avg/max/mdev = 0.385/0.547/0.676/0.103 ms
```

2.3 Connectivity test for Server 1 to Server 2: / Successful ☐ Not Successful

```
vboxuser@JeraldServer1:~$ ping 192.168.56.115
PING 192.168.56.115 (192.168.56.115) 56(84) bytes of data.
64 bytes from 192.168.56.115: icmp_seq=1 ttl=64 time=0.732 ms
64 bytes from 192.168.56.115: icmp_seq=2 ttl=64 time=0.479 ms
64 bytes from 192.168.56.115: icmp_seq=3 ttl=64 time=0.433 ms
64 bytes from 192.168.56.115: icmp_seq=4 ttl=64 time=0.437 ms
64 bytes from 192.168.56.115: icmp_seq=5 ttl=64 time=0.494 ms
^C
--- 192.168.56.115 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4101ms
rtt min/avg/max/mdev = 0.433/0.515/0.732/0.111 ms
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 `ssh username@ip_address_server1` for example, `ssh jvtaylor@192.168.56.120`

1.2 Enter the password for server 1 when prompted

1.3 Verify that you are in server 1. The user should be in this format `user@server1`. For example, `jvtaylor@server1`

2. Logout of Server 1 by issuing the command `control + D`.

```
vboxuser@workstation:~$ ssh vboxuser@192.168.56.114
The authenticity of host '192.168.56.114 (192.168.56.114)' can't be established.
ED25519 key fingerprint is SHA256:9YEPdI6+HHX6dlc1Wpk1mXNp6k10fYqYm7G8XDeQRUU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.114' (ED25519) to the list of known hosts
.
vboxuser@192.168.56.114's password:
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 6.8.0-124-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

   https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

vboxuser@JeraldServer1:~$
logout
Connection to 192.168.56.114 closed.
```

3. Do the same for Server 2.


```
vboxuser@workstation:~$ ssh vboxuser@192.168.56.115
The authenticity of host '192.168.56.115 (192.168.56.115)' can't be established.
ED25519 key fingerprint is SHA256:MhCpBbJn7RlDMbE3u//qIwtixkb6iCAxvq4AD5SG+Is.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.115' (ED25519) to the list of known hosts
.
vboxuser@192.168.56.115's password:
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

   https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.
```

```
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

vboxuser@Jeraldserver2:~$
logout
Connection to 192.168.56.115 closed.
```

4. Edit the hosts of the Local Machine by issuing the command **sudo nano /etc/hosts**. Below all texts type the following:
 - 4.1 **IP_address server 1** (provide the ip address of server 1 followed by the hostname)
 - 4.2 **IP_address server 2** (provide the ip address of server 2 followed by the hostname)
 - 4.3 Save the file and exit.

```

vboxuser@workstation: ~
GNU nano 6.2 /etc/hosts *
127.0.0.1    localhost
127.0.0.1    workstation
192.168.56.114 server1
192.168.56.115 server2

# The following lines are desirable for IPv6 capable hosts
::1        ip6-localhost ip6-loopback
fe00::0    ip6-localnet
ff00::0    ip6-mcastprefix
ff02::1    ip6-allnodes
ff02::2    ip6-allrouters
  
```

[^]G Help [^]O Write Out [^]W Where Is [^]K Cut [^]T Execute [^]C Location
[^]X Exit [^]R Read File [^]Y Replace [^]U Paste [^]J Justify [^]_ Go To Line

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do *ssh jvtaylor@server1*. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Server1

```
vboxuser@workstation:~$ ssh vboxuser@server1
The authenticity of host 'server1 (192.168.56.114)' can't be established.
ED25519 key fingerprint is SHA256:9YEPdI6+HHX6dlc1Wpk1mXNp6k10fYqYm7G8XDeQRUU.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
vboxuser@server1's password:
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 6.8.0-124-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
  Ubuntu Pro subscription. Free for personal use.

  https://ubuntu.com/pro
```

```
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
```

```
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
  Ubuntu Pro subscription. Free for personal use.

  https://ubuntu.com/pro
```

```
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

```
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings
```

```
Last login: Thu Jul 16 08:59:31 2026 from 192.168.56.113
```

```
vboxuser@JeraldServer1:~$
```

```
logout
```

```
Connection to server1 closed.
```

Server2

```
vboxuser@workstation:~$ ssh vboxuser@server2
The authenticity of host 'server2 (192.168.56.115)' can't be established.
ED25519 key fingerprint is SHA256:MhCpBbJn7RlDMbE3u//qIwtixkb6iCAxvq4AD5SG+Is.
This host key is known by the following other names/addresses:
```

```
  ~/.ssh/known_hosts:4: [hashed name]
```

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
```

```
vboxuser@server2's password:
```

```
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 6.8.0-134-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage
```

```
* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
  Ubuntu Pro subscription. Free for personal use.
```

```
https://ubuntu.com/pro
```

```
Expanded Security Maintenance for Applications is not enabled.
```

```
0 updates can be applied immediately.
```

```
* Support:        https://ubuntu.com/advantage
```

```
* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
  Ubuntu Pro subscription. Free for personal use.
```

```
https://ubuntu.com/pro
```

```
Expanded Security Maintenance for Applications is not enabled.
```

```
0 updates can be applied immediately.
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

```
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings
```

```
Last login: Thu Jul 16 08:47:57 2026 from 192.168.56.113
```

```
vboxuser@Jeraldserver2:~$
```

```
Logout
```

```
Connection to server2 closed.
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. How are we able to use the hostname instead of IP address in SSH commands?
 - I used a hostname like the server1 in the SSH command. My computer translates it into an IP address using a local lookup file called `/etc/hosts`, which acts like a private digital phonelook. When you edit this file to map a name directly to a specific IP address (such as 192.168.56.114), your system reads this mapping first and immediately routes the SSH traffic to the correct

destination, completely bypassing the need to query internet-based DNS servers.

2. How secured is SSH?

- SSH (Secure Shell) is exceptionally secure because it completely encrypts all data, including passwords, commands, and files, transmitted between your workstation and the remote server, leaving network eavesdroppers with nothing but unreadable, scrambled text. Furthermore, it prevents data tampering in transit by automatically dropping connections if any modification is detected, and it supports virtually unhackable cryptographic key pairs that completely eliminate the risk of password brute-force attacks.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini and QuillBot

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Grammar correction and explanation of the concepts and functionality.

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.